Dario Loi

Boston, MA | dario-loi.github.io | github.com/dario-loi | Google Scholar

Research Interests: Topological deep learning, sheaf neural networks, graph neural networks, simplicial complexes

EDUCATION

Northeastern University September 2026 – Present
Ph.D. in Computer Engineering, MENTIS Lab, advised by Prof. Francesco Restuccia *Boston, MA*

Sapienza University of Rome September 2023 – October 2025
M.S. in Computer Science *Rome, Italy*
GPA 29.8/30 (3.98/4.0), 110/110 cum laude; Honors Program with Prof. Iacopo Masi

Sapienza University of Rome September 2020 – July 2023
B.S. in Applied Computer Science and Artificial Intelligence *Rome, Italy*
GPA 28.4/30 (3.79/4.0), 110/110 cum laude; Honors Program with Prof. Danilo Avola

PUBLICATIONS

Implicit Inversion turns CLIP into a Decoder
A. D'Orazio, M.R. Briglia, D. Crisostomi, **D. Loi**, E. Rodolà, I. Masi
International Conference on Learning Representations (ICLR), 2026 (poster)

- Ran the hyperparameter searches and produced qualitative results

A Metacognitive Framework for Test-Time Adaptation: Taxonomy and Research Challenges
D. Loi, G. Onorato, A. Coviello, S. Rifat, N.D. Bastian, F. Restuccia
Proceedings of the AAAI Symposium Series, vol. 10(1), pp. 147–162 (METACOG-26), 2026. Top 3 of 27 papers

- Built the TTA taxonomy; ran the experiments in which deliberate cues admitted up to 22.1x fewer OOD samples

Implicit Sheaf Operators for Effective and Efficient Cellular Sheaf Learning
G. Onorato, **D. Loi**, C. Apostol, N.D. Bastian, F. Restuccia
Preprint, 2026

- Designed Sheaf Convolutional Networks (up to +3.93% accuracy over NSD) and proved complexity bounds

Topological Deep Learning Challenge 2026: Bridging the Gap
G. Bernárdez, L. Telyatnikov, M. Papillon, ..., **D. Loi**, G. Onorato, ..., N. Miolane
PMLR, vol. 334, pp. 4–29 (TAG-DS 2026), 2026. Our entry won 1st of 37 in Track 1 (GNNs)

- Implemented the Track 1 winning entry, Directed Sheaf NNs, and four more sheaf models (five pull requests)

Topological Deep Learning Challenge 2025: Expanding the Data Landscape
G. Bernárdez, L. Telyatnikov, M. Papillon, ..., **D. Loi**, ..., N. Miolane
Proceedings of Machine Learning Research, vol. 321, pp. 4–14 (TAG-DS 2025), 2026

- Contributed a TopoBench loader for MoleculeNet datasets

AutoBench: Automating LLM Evaluation through Reciprocal Peer Assessment
D. Loi, E.M. Muià, F. Siciliano, G. Trappolini, V. Crisà, P. Kruger, F. Silvestri
Preprint, arXiv:2510.22593, 2025

A Metacognitive Perspective on Anomaly Detection with Vision Foundation Models
M. Galiazzo, G. Onorato, **D. Loi**, N.D. Bastian, F. Restuccia
Proceedings of the AAAI Symposium Series, vol. 10(1), pp. 110–114 (METACOG-26), 2026

- Supervised the first author's thesis research

EXPERIENCE

Research Intern December 2025 – June 2026
Evidence Srl *Pisa, Italy*

- Prototyped an agentic LLM for Simulink-style power plant control, scaling reliable output from 5 to 100+ blocks
- Orchestrated it with vLLM, LangChain, and LangGraph, using structured outputs and a custom DSL for valid outputs

Honors Program Researcher, M.S. January 2025 – October 2025
Sapienza University of Rome *Rome, Italy*

- Honors Program research with Prof. Iacopo Masi

Teaching Assistant May 2024 – September 2024
Computer Architecture I and II, B.S. in Computer Science, Sapienza University of Rome *Rome, Italy*

- Taught logic gates, MUX and registers, counters, RISC-V, pipelines, caches, and custom instruction design
- Administered exams, homework, and practice sessions, and held 1-on-1 sessions with students

Computer Vision Software Engineer October 2023 – June 2024
Sapienza Flight Team *Rome, Italy*

- Built real-time YOLOv8 detection of payload drop sites from a drone in flight (SUAS 2024)

Honors Program Researcher, B.S. March 2022 – July 2023
Sapienza University of Rome *Rome, Italy*

- Worked on AI methods for motion video enhancement, supervised by Prof. Danilo Avola

HONORS AND AWARDS

- 1st of 37 in Track 1 and Excellent Portfolio honorable mention, Topological Deep Learning Challenge 2026
- Top 3 of 27 papers, AAAI Fall Symposium on Artificial Metacognition (METACOG-26) 2026
- One of 10 students selected across Sapienza's I3S faculty, Xi'an Jiaotong University summer camp 2025
- Honors Program, Sapienza University of Rome: B.S. (one of eight selected) and M.S. 2022 – 2025

RESEARCH SOFTWARE AND OPEN SOURCE

TopoBench (Topological Deep Learning Challenges 2025 and 2026) github.com/geometric-intelligence/topobench

- Implemented the winning GNN-track entry, Directed Sheaf NNs (Fiorini et al., 2026), and four more sheaf models
- Co-authored the peer-reviewed challenge reports at TAG-DS 2025 and 2026 (PMLR)

Graph Neural Networks for Next-POI Recommendation (M.S. Project) github.com/dario-loi/foursquare

- Designed a transformer-based GNN in PyTorch Lightning: +2.3% top-1, +13.7% top-20 accuracy over prior methods

StapleGL (51 GitHub stars) github.com/dario-loi/staplegl

- Header-only, dependency-free OpenGL wrapper in modern C++20

StemSep (23 GitHub stars, 5 forks) github.com/dario-loi/stemsep

- Web app that splits audio into stems with Demucs behind a Gradio interface, shipped as a Docker image

CONFERENCES, PRESENTATIONS, AND SCHOOLS

- AAAI Fall Symposium on Artificial Metacognition (METACOG-26) (presented TTA paper) November 2026
- FUSION 2026, Trondheim, Norway (presented Decision-Driven Credal Information Fusion for the authors) June 2026
- ICLR 2026, Rio de Janeiro, Brazil (poster presentation: Implicit Inversion turns CLIP into a Decoder) April 2026
- ROCKIN' AI 2.0, Roccella Conference on Inference and AI, Roccella Ionica, Italy September 2025
- Oxford Machine Learning Summer School (OxML), Representation Learning and Generative AI August 2025
- IT summer camp, Xi'an Jiaotong University, China (one of 10 selected at Sapienza I3S) July – August 2025

SERVICE AND LEADERSHIP

Department Student Representative 2020 – 2023
Computer Science Department, Sapienza University of Rome *Rome, Italy*

Faculty Student Representative 2022 – 2025
Sapienza University of Rome *Rome, Italy*

Reviewer: ICLR, TMLR, KDD, ECML-PKDD, WWW, INFOCOM, MILCOM, AAAI, CIKM, TKDD
Mentoring: Supervised a thesis student's anomaly detection research (Galiazzo et al., METACOG-26)

SKILLS

Programming: Python, C++, C, CUDA, Java, SQL; Git, Linux, Docker, SLURM, Weights & Biases
Machine Learning: PyTorch, PyTorch Geometric, TopoX, Hugging Face, LangGraph, LangChain, vLLM